WEB APPLICATION ARCHITECTURE

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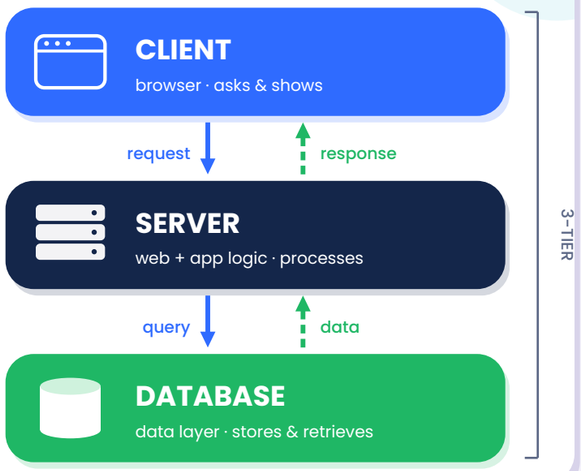
"What happens behind the screen when a user interacts with a web application?"

1 What is Web App Architecture?

The blueprint of how a web app is organised: the browser, the server, the database and the security that links them. Together they turn one click into a result on your screen.

Client · Server · Database · Security

2 Client–Server Architecture



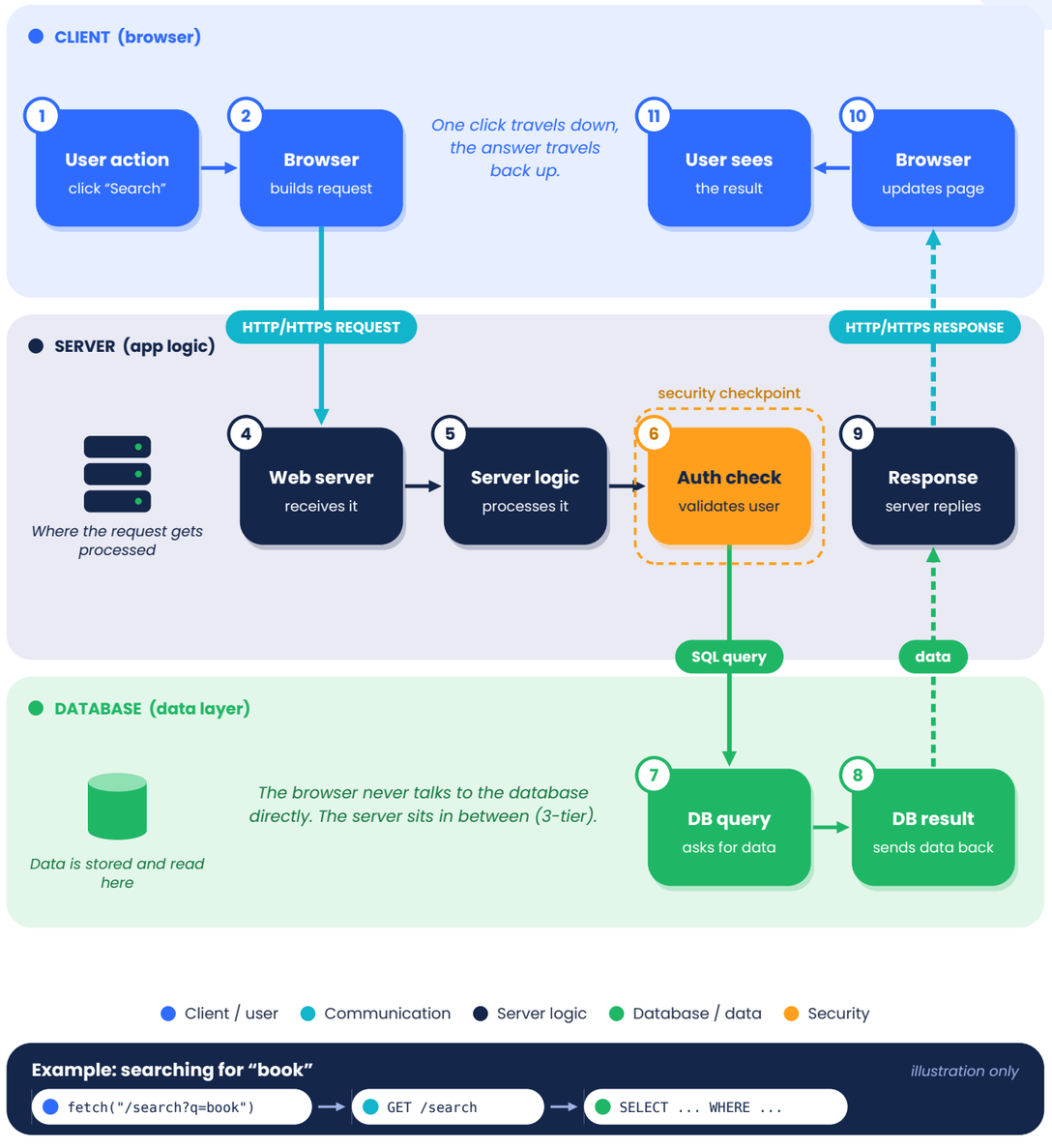
3 HTTP vs HTTPS

HTTP
The rules for sending requests and responses

HTTPS
HTTP + TLS encryption, so data in transit is protected

HTTPS protects data in transit only. Apps still need validation, login and safe sessions.

7 Complete Web Application Flow



4 Request & Response

REQUEST *browser to server*

```
GET /search?q=book HTTP/1.1
```

Method what to do: GET, POST
URL where to go + query
Headers cookies, browser, type
Body data sent (forms)

RESPONSE *server to browser*

```
HTTP/1.1 200 OK
```

Status the result code
Headers type, cookies
Body HTML, JSON, images

200 OK · 301 Moved · 404 Not found · 500 Error

GET read · POST send · PUT update · DELETE remove

5 Workflow & Real World

BUILD WORKFLOW

Plan → Design UI → Build client → Events → Server logic → Database → Security → Test → Deploy

APP LIFE CYCLE

Design → Development → Testing → Deployment → Maintenance

SAME ARCHITECTURE, DIFFERENT APPS

Banking · E-commerce · College portal · Healthcare · Social media · Learning · Booking

6 Database Connectivity

App → Connection → Query → Result

id	task
101	view creation
102	view cursor

```
CREATE VIEW v101 AS
SELECT name FROM Students;
DECLARE c102 CURSOR FOR
SELECT * FROM v101;
```

SQL talks to relational databases

8 Conclusion

You click, the browser asks, the server thinks, the database gives data, and you see the answer. That is how a web app works.

THE JOURNEY OF A CLICK

USER → REQUEST → PROCESS → DATA → SECURITY → RESPONSE → USER

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